

Vulnerability Assessment of Nitrate Contamination in the Gulf Coast Aquifer, Texas

A Proactive Hydro-Geospatial & Socioeconomic Analysis

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The Silent Crisis

Fragile Resource

Groundwater remains a fragile resource for rural populations. The highly soluble nature of nitrate-N makes contamination silent and physically unregulated in domestic wells.



Methodological Integration

Predictive Synthesis

Synthesizing geospatial land-use metrics, Multiple Logistic Regression (MLR), and Life Cycle Costing (LCC) to shift from reactive to predictive analysis.



Proactive Prioritization

Diagnostic Power

Statistical vulnerability assessments provide strong diagnostic power (AUC 0.74–0.75), effectively replacing cost-prohibitive physical testing to target vulnerable wellheads.

Layer 1:
Surface / Root Zone

Layer 2:
Vadose Zone

Layer 3:
Saturated Thickness
/ Aquifer

High solubility bypasses
natural uptake.

Leaching
Corridor

The Reversibility Problem

Once contaminated, these hydrogeological systems require **decades of natural attenuation** or complex, cost-prohibitive technical intervention.

Primary Sources of Nitrate Pollution

Fertilizers



Direct leaching of excess NO_3^- ; conversion of NH_4^+ into mobile mobile nitrate.



Manure



Organic nitrogen mineralized to NO_3^- ; non-point source leaching from livestock.



Industrial



Point source discharge and chemical runoff leaching into the water table.



Landfill



Pollution via leachate migration through compromised liners.



Residential



Leaking septic systems and turf-grass fertilizer leaching.



Geographic Context

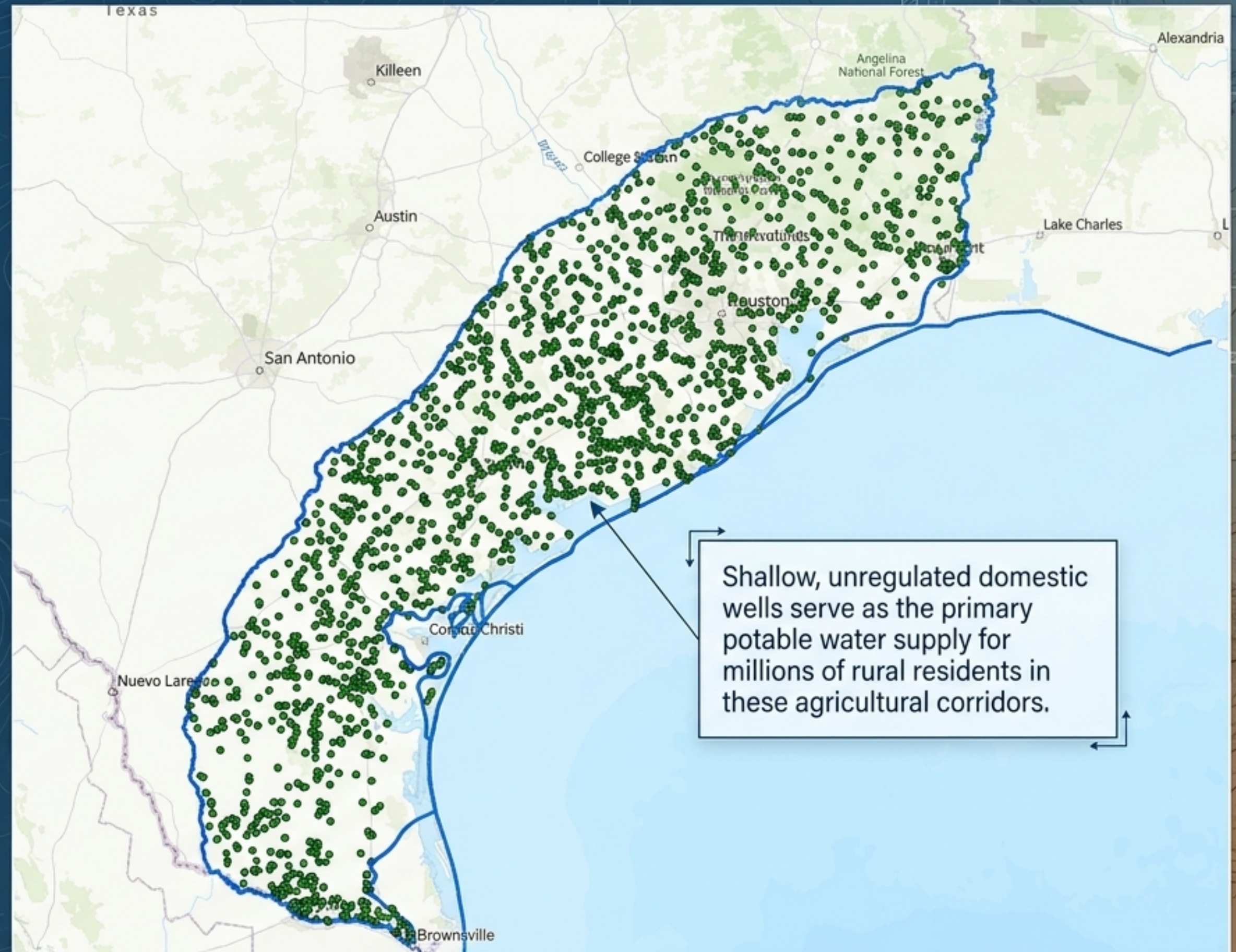
Data Source: TWDB
Groundwater Database
(GWDB)

Total Observations: 2,276
Wells initially.

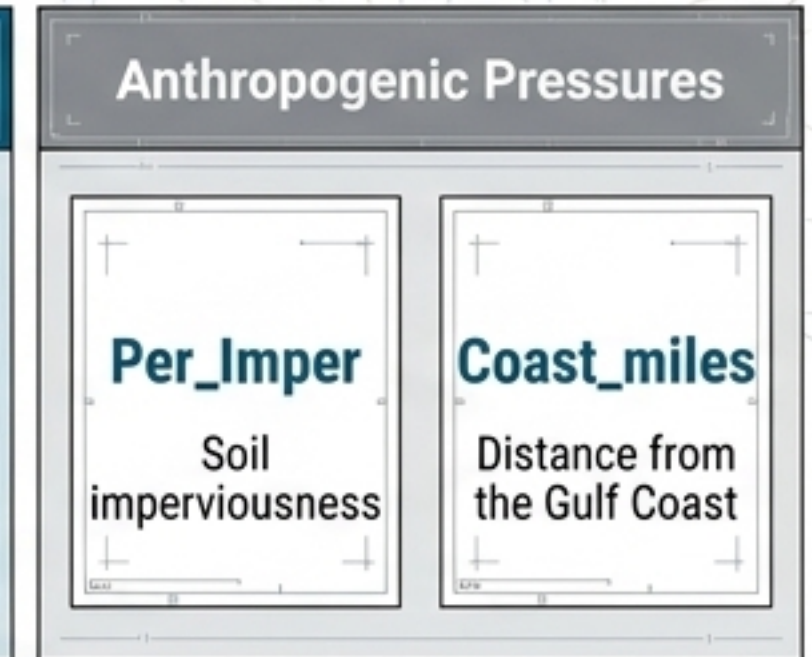
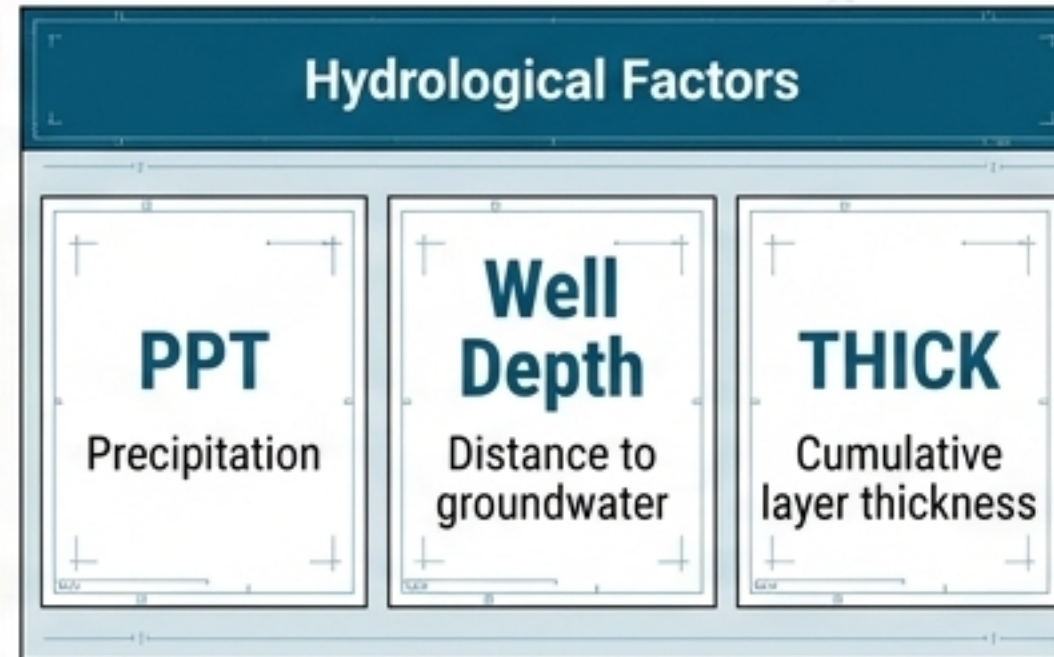
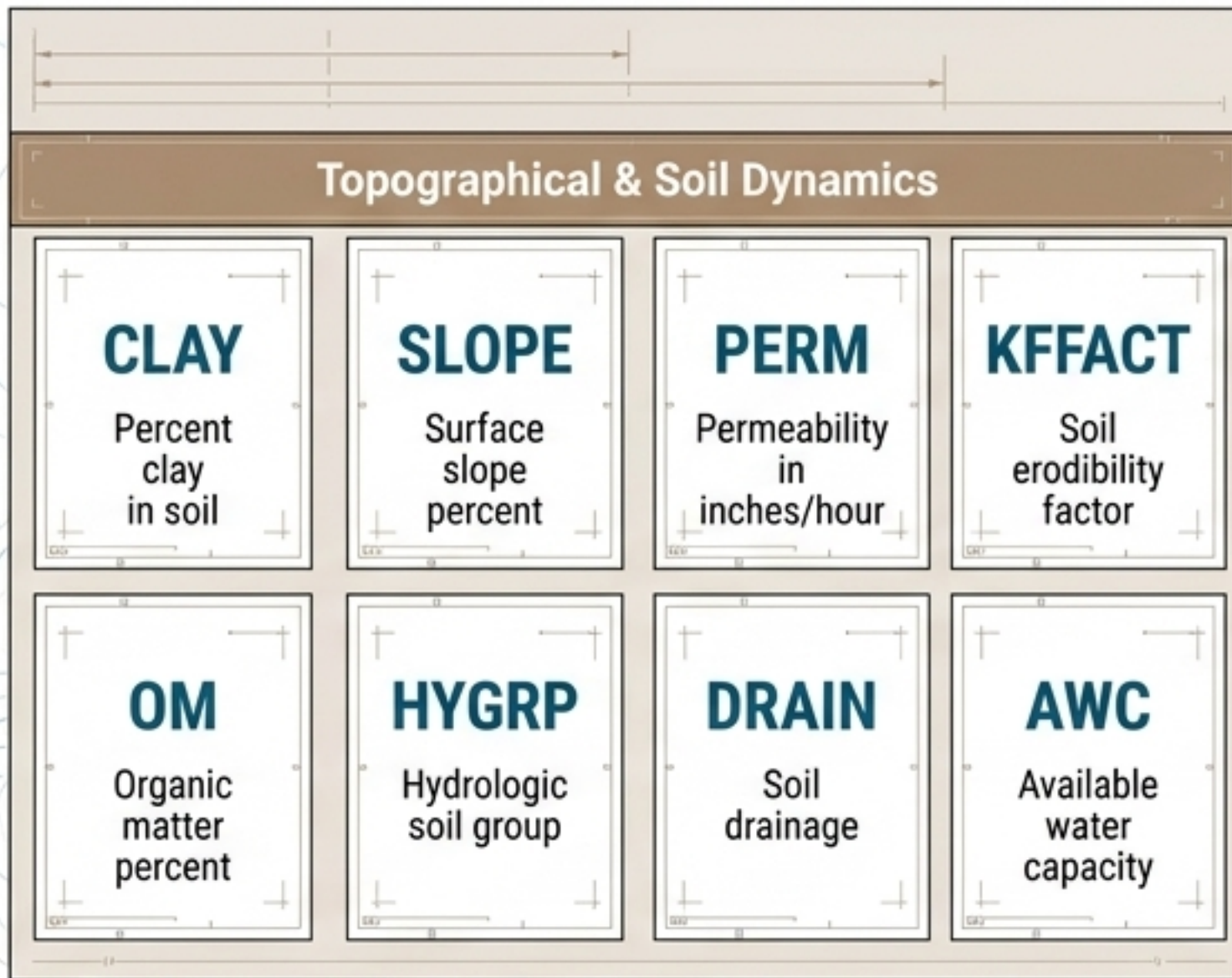
Risk Filter: Filtered for wells
< 1,000 feet depth.

1,889

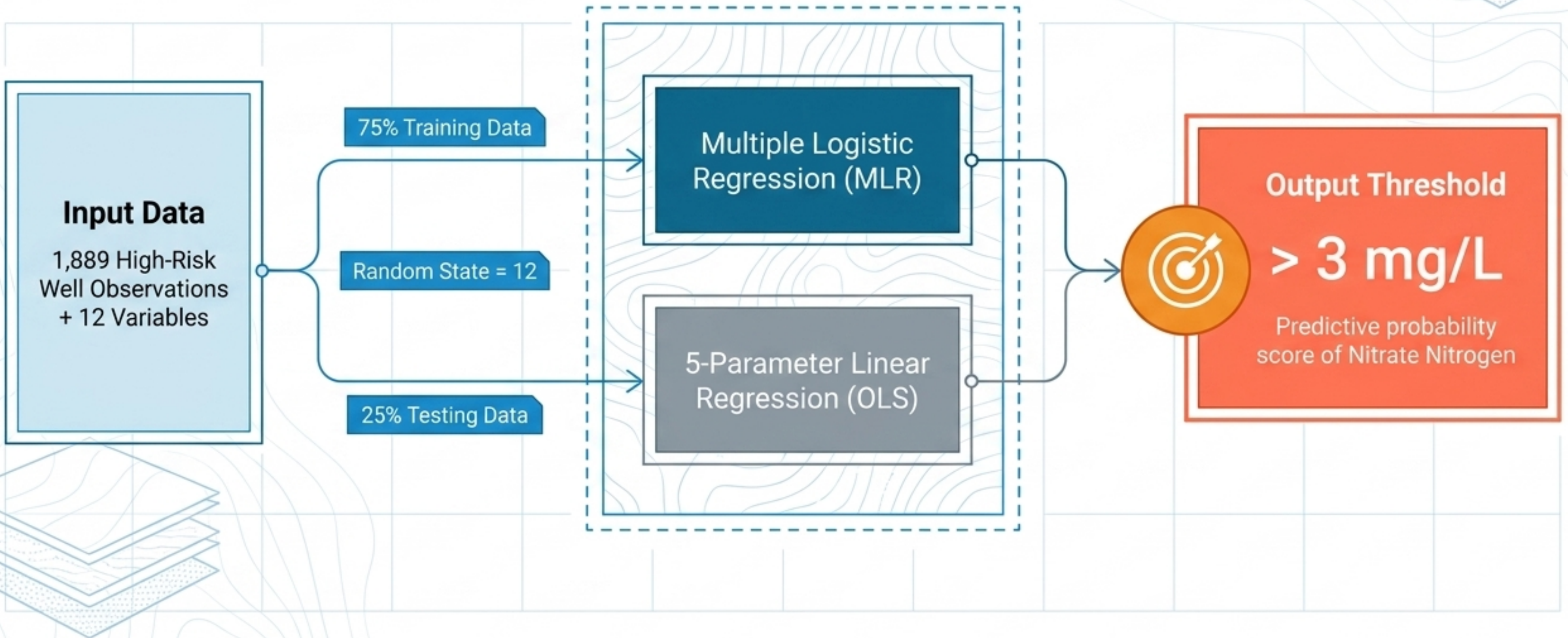
high-risk observations
utilized for modeling.



The Data Architecture: 12 Independent Variables



Methodology: The Predictive Engine



Model Diagnostics: The Limits of Linear Regression

OLS Regression Results						
=====						
Dep. Variable:	Avg_Nit	R-squared:	0.175			
Model:	OLS	Adj. R-squared:	0.172			
Method:	Least Squares	F-statistic:	59.06			
Date:	Thu, 02 Mar 2023	Prob (F-statistic):	7.31e-56			
Time:	05:31:42	Log-Likelihood:	-5832.8			
No. Observations:	1397	AIC:	1.168e+04			
Df Residuals:	1391	BIC:	1.171e+04			
Df Model:	5					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	32.3115	3.154	10.246	0.000	26.125	38.498
PPT	-0.0140	0.002	-7.151	0.000	-0.018	-0.010
Coast_miles	0.0628	0.020	3.154	0.002	0.024	0.102
SLOPE	-1.4615	0.411	-3.557	0.000	-2.268	-0.655
DRAIN	-2.3030	0.804	-2.864	0.004	-3.880	-0.726
Well Depth	-0.0095	0.002	-5.600	0.000	-0.013	-0.006
=====						
Omnibus:	1856.576	Durbin-Watson:	2.021			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	427677.172			
Skew:	7.155	Prob(JB):	0.00			
Kurtosis:	87.514	Cond. No.	8.81e+03			
=====						

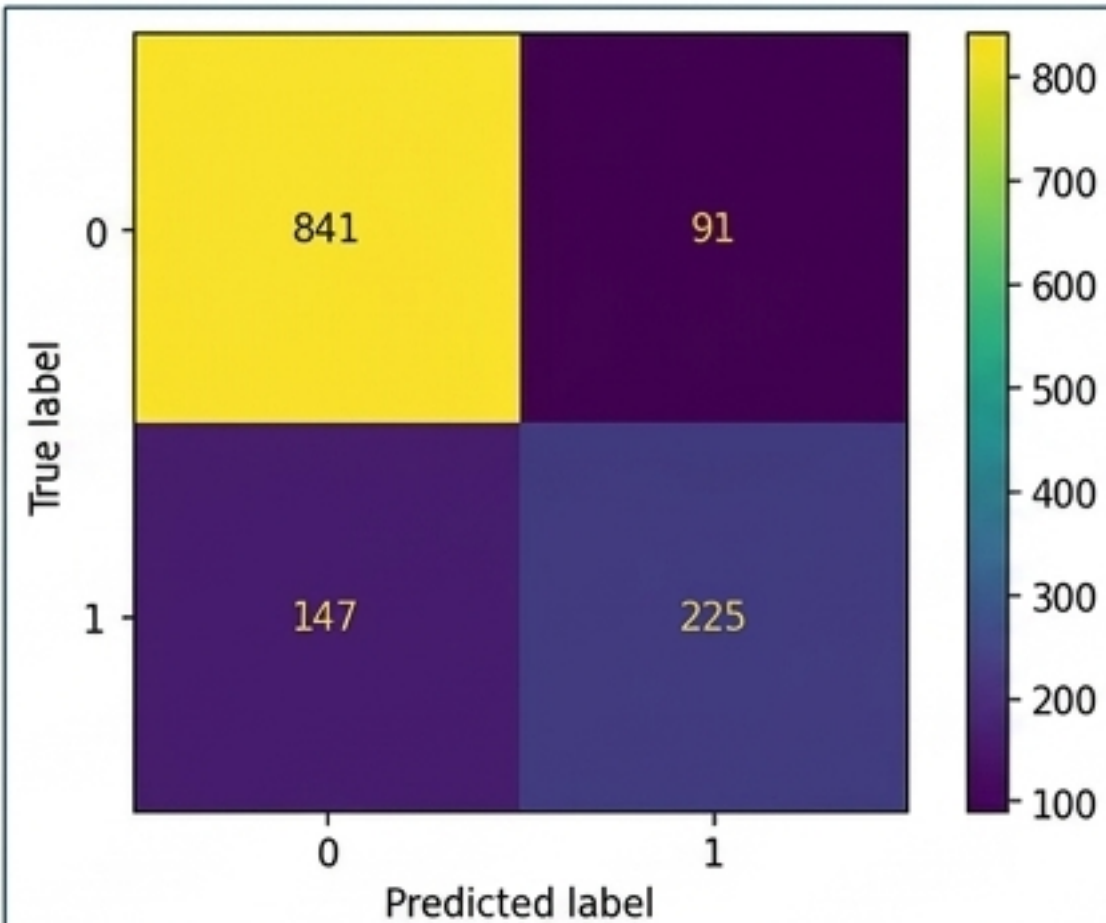
Variables encompass 0 in the 95% confidence limit, rendering them statistically unreliable.

Multiple parameters exhibit $P > 0.05$, failing the threshold for statistical significance.

Result: Models 1 & 3 (5-Parameter Linear Regression) demonstrate lower false positives, but fail to capture the true state of vulnerability with sufficient statistical rigor.

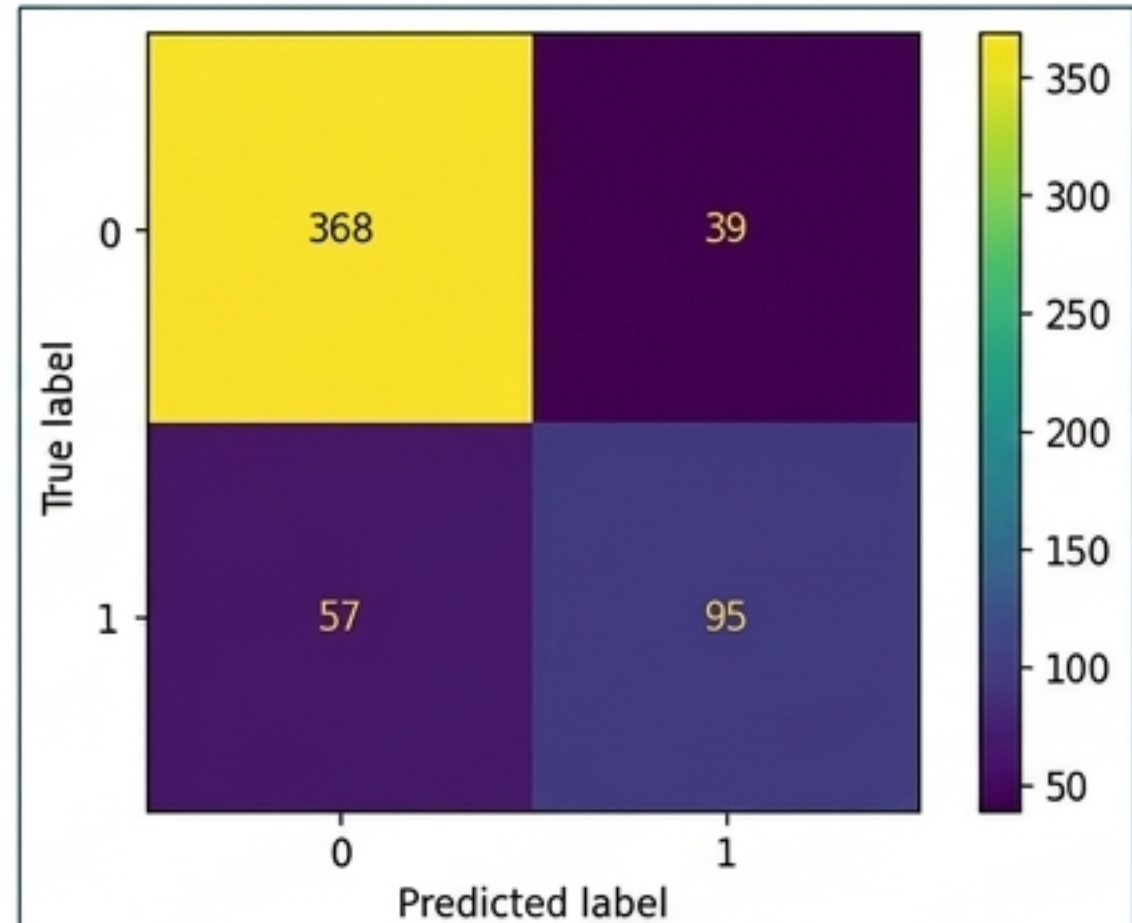
Logistic Regression: Superior Predictive Power

Training Data - Model 2



81.7% Accuracy

Testing Data - Model 2

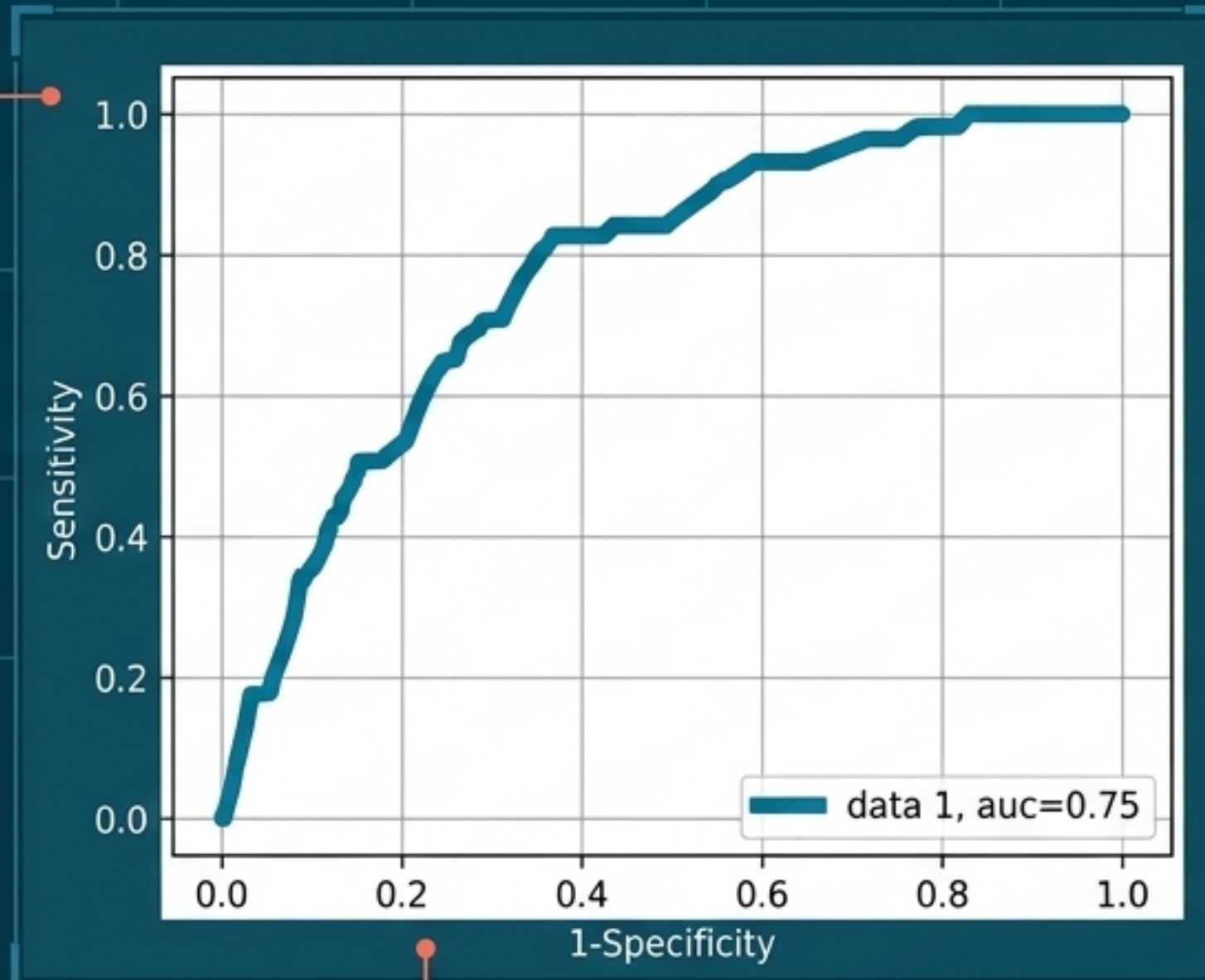


82.8% Accuracy

Unlike linear modeling, Multiple Logistic Regression maintains robust consistency between training and testing data, correctly predicting the binary state of wellhead contamination (> 3 mg/L).

Validation: Diagnostic Power

True Positive Rate:
The model's ability
to accurately
identify genuinely
contaminated wells.



False Positive Rate: The
model's ability to avoid
triggering false alarms for
clean wells.

**AUC:
0.74 – 0.75**

An Area Under the Curve (AUC) of 0.75 signifies strong diagnostic power. This model can reliably prioritize public health interventions without the financial waste of blanket physical testing.

The Primary Predictors of Vulnerability



Shallow Architecture

Well Depth

Wells shallower than 1,000 feet exhibit the highest vulnerability, serving as the primary corridor for surface-derived nitrate.



Hydrological Forcing

Precipitation (PPT)

High rainfall acts as the primary transport mechanism, actively leaching soluble nitrates through the vadose zone.



Anthropogenic Capping

Imperviousness & Coast Proximity

High surface imperviousness dictates the rate of recharge and concentrates runoff pathways, accelerating contamination in clustered zones.

Socioeconomic Synthesis: Life Cycle Costing (LCC)

Remediation strategies based on the 10 mg/L legal threshold.

RO Point-of-Use

Household Level (3-4 pax)



**\$4 – \$164
/ year**

Highly cost-effective for individual rural households. Lowest initial footprint.

Ion Exchange

Household Level



**Mid-Tier
Cost**

Viable secondary option for individual domestic application where RO is unavailable.

RO Point-of-Entry

Community Level



**Highest
Cost**

Capital-intensive initial setup with extremely high ongoing operation & maintenance costs.

Bio-denitrification

Community Level (~10k pax)



**Most Efficient
Scaled Cost**

Least expensive community option due to tremendous operational economies of scale.

Strategic Conclusion: From Reactive to Proactive

The Reactive Past

Reactive Management



Relies entirely on **physical testing** of individual wells.

Logistically slow, geographically limited, and deeply **cost-prohibitive** at scale.

Intervention only occurs after contamination exceeds the dangerous 10 mg/L MCL threshold.

The Proactive Future

Statistical Vulnerability Assessment

MLR modeling paired with geospatial land-use data.

Identifies **at-risk well clusters** before they breach public health thresholds.

Directs precise, highly cost-effective deployment of **targeted POU/POE treatment systems** based on life-cycle socioeconomics.

References



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Keeler, B. L., & Polasky, S. (2014). Land-use change and costs to rural households: A case study in groundwater nitrate contamination. *Environmental Research Letters*.



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